

PAPER_1392 — UQFF Resolution of the Renninger Negative-Result Experiment (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Quantum Measurement (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "renninger_negative_result"})` **Calculator surface:** `calculate_paradox({"paradox": "renninger_negative_result"})` **Closure helper:** `_l96_uqff_axiom_renninger_negative_result_closure()`

The Paradox

A wavefunction split into branches by a partial-coverage detector. If the detector does NOT click in time T, the wavefunction collapses into the un-detected branches — a ‘collapse without measurement event’ apparent paradox.

UQFF Closed Identity

$F_U = 1$ redistribution fraction = 1.0 EXACT among remaining branches

$F_U = 1$ global ledger normalization: at the no-click time T, the $F_U = 1$ amplitude redistributes 100% into the remaining branches. The ‘collapse’ is the F_U renormalization event, not an interaction.

Physical Interpretation

Renninger’s paradox is resolved by recognizing that the $F_U = 1$ ledger normalization is the mechanism of ‘collapse’ — no particle-detector interaction is needed because F_U redistribution is a ledger event, not a physical scattering.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "renninger_negative_result"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_renninger_negative_result_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` $\rightarrow$ `_l96_uqff_axiom_renninger_negative_result_closure`
- Dispatch keys: `renninger_negative_result`, `renninger`
- Related foundational papers: PAPER_646 ($F_U B_i / F_U=1$ ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF dispatch closure for the Renninger Negative-Result Experiment via the identity above.